



Variable stiffness of multi-teeth mechanical metamaterials

Xin Liu ^a, Shuai Chen ^{a,b}, Bing Wang ^{a,*}, Xiaojun Tan ^{c,*}

^a National Key Laboratory of Science and Technology on Advanced Composites in Special Environments, Harbin Institute of Technology, Harbin, 150080, PR China

^b Zhengzhou Research Institute, Harbin Institute of Technology, Zhengzhou, 450046, PR China

^c School of Civil Aviation, Northwestern Polytechnical University, Xi'an, 710072, PR China

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ABSTRACT

To address the growing demands for high-performance, multifunctional, and dynamically adaptive systems, variable stiffness structures have emerged as a transformative design paradigm. In this paper, a groundbreaking multi-teeth (MT) mechanical metamaterial (MM) is proposed, providing excellent tunable stiffness properties and rich stiffness variations. It could achieve precise stiffness decoupling and control through rotational modulation, allowing transitions from extremely flexible to rigid states. The MT-MM's unique architecture offers transformative potential for applications in flexible robotic joints, adaptive aircraft landing gears, etc., offering a brand new solution and reference for researchers in the field of variable stiffness.

1. Introduction

With the evolution of science and technology, structures are exposed to ever more complex and diverse working environments and task requirements [1,2]. Traditional fixed-stiffness structures are difficult to meet the demands for high performance, multi-functionality and dynamic adaptation in various cases [3–6]. Variable stiffness structures, as an innovative structural design concept, are emerging as a research hotspot, demonstrating unique advantages and great application potential in numerous fields [7–9]. It can adjust its stiffness characteristics in real time according to different loading conditions. For example, the stiffness could be improved to ensure the strength and stability of the structure under larger loads [10–12], while the stiffness might be appropriately reduced if the loads are small, so as to avoid stress concentration or other unfavourable effects caused by the excessive stiffness, improving the reliability and durability in the whole service life. Additionally, variable stiffness structures could play a key role in energy transmission and conversion systems, such as the joint drive of flexible robots [13,14]. By adjusting the stiffness, the path and approach of force transmission could be optimized, leading to a reduction of energy loss in the process.

Due to high load bearing capacity and rich mechanical response, conical shell structure has been attracting much attention [15,16]. In the field of MM, since the shell would perform elastic jump under compressive loading, it can demonstrate rich stiffnesses, such as positive, negative and quasi-zero-stiffness through the adjustment of structural parameters [17,18]. However, this also causes some problems concurrently. As the elastic jump behaviour of conical shell structures requires

large deformations to occur, researchers usually have no choice but to adopt some flexible materials for preparation, such as silicone, polyurethane elastic rubber, etc., ensuring a favourable reusability of the structure. It is definitely at the expense of the load-bearing ability. Additionally, traditional truncated conical shell structures are difficult to modify their mechanical properties after preparation. Therefore, how to design variable stiffness structures with specific load-bearing capacity is a challenging issue to be addressed urgently.

In this paper, a MT-MM with variable stiffness characteristics is proposed by introducing a rotational adjustment. Compared to conventional truncated conical shell structures, it no longer has only one mechanical property, achieving stiffness decoupling with superior adjustability. Through a combination of numerical simulation and experiment, this paper comprehensively investigates the influence of structural parameters and arrangement on the mechanical properties of MT-MM. The article is organized as follows: Firstly, the structural form of MT-MM is described and the finite element numerical model is established, as well as the experimental method of quasi-static compression. Secondly, the influence of structural parameters on the compression characteristics of MT-MM is thoroughly investigated, verifying the adjustable stiffness ability of unit cells. Finally, the mechanical property variations of metamaterials under different arrangements are explored. The MT-MM proposed in this paper provide a novel idea for the research in the field of variable stiffness.

* Corresponding authors.

E-mail addresses: wangbing86@hit.edu.cn (B. Wang), xiaojun.tan@nwpu.edu.cn (X. Tan).

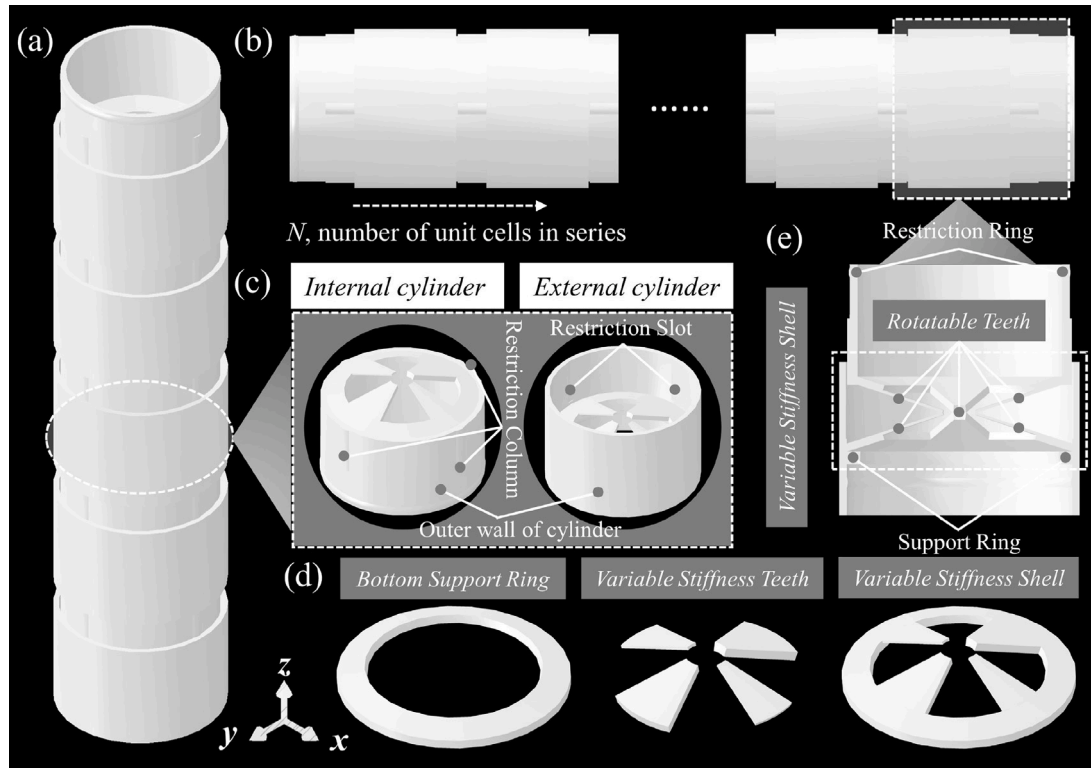


Fig. 1. (a)–(b) Schematic diagram of overall MT-MM. (c) Internal and external cylinders of the metamaterial. (d) Composition of the variable stiffness shell. (e) Unit cell of the MT-MM.

2. Design philosophy of structures and methods

2.1. Structural form and design

To achieve stiffness decoupling of truncated conical shell structures, a MT-MM is presented in this paper. As shown in Fig. 1(a)–(b), the global model of MT-MM is demonstrated, defining the number of unit cells as N and the number of teeth pairs as N_p . As illustrated in Fig. 1(c), the unit cell of MT structure is divided into internal and external cylinders. The term ‘MT’ refers to the geometric configuration of the unit cell, consisting of several inclined teeth. By adjusting the contact conditions between different teeth, it could generate different stiffness characteristics. Meanwhile, due to the small contact between different teeth during compression, in order to limit the displacement of the structure, a restriction column is attached to the internal cylinder, while a restriction slot is matched and designed on the external cylinder. Particularly, the limiting portion could also serve as a precise positioning upon the rotational adjustment of the structural stiffness. During rotation, the MT-MM could only be compressed while the restriction columns align with the slots. Otherwise, it would be more difficult to deform due to friction. The number and distribution of the restriction columns and slots vary with the number of N_p and the angle is calculated as $180^\circ/N_p$. As depicted in Fig. 1(d), the variable stiffness shell is the main bearing structure of internal and external cylinders, consisting of the bottom support ring and the variable stiffness MT. As indicated in Fig. 1(e), for the convenience of structural assembly, a restriction ring and a support ring are specially arranged on the internal and external cylinders, in which the restriction ring plays the role of ‘clamping’ during the connection of neighbouring internal and external cylinders, preventing the detachment of different unit cells. The support ring serves to convey the force between the cells. The stiffness of the metamaterial could be adjusted through rotating a certain angle to regulate the contact of different teeth. It achieves tunable stiffness through a unique geometric configuration, where the

interaction between different teeth enables precise stiffness modulation. By rotating the unit cells, the contact conditions of teeth would change, leading to variations in stiffness. This mechanism allows the structure to transition from extremely flexible to rigid states, providing exceptional adaptability to different loading conditions.

2.2. Finite element model

Through the commercial finite element software ABAQUS, numerical simulations of the MT-MM are carried out. As displayed in Fig. 2(a), C3D4 elements are adopted to mesh the internal and external cylinders, while the hexahedral grids C3D8R are applied to the main bearing structures. The different elements in the contact region are bounded by the ‘Tie’ constraints in the interaction module. After the mesh convergence analysis, the total mesh size of the tetrahedral elements is set to 2 mm, and that of the hexahedral elements is set to 1 mm. A reference point is defined directly above the structure, and the it is constrained to the upper surface of the metamaterial through ‘coupling’ in the interaction module. Meanwhile, the displacement and reaction force data of the reference point are obtained in the history output. The material parameters of the metamaterial are set as follows: modulus of elasticity 2136 MPa, Poisson’s ratio 0.4 and density 1180 kg/m³ (provided by Beijing YiSuPuRui Technology Co., Ltd.).

2.3. Quasi-static compression experiment

In this paper, additive manufacturing technology SLA (Stereo Lithography Appearance) is adopted for structure preparation. It utilizes specific intensity laser focusing irradiation on the surface of light-curing material (mainly photosensitive resin) to complete the printing of single layer from point to line and line to surface, and then realizes the preparation of the complete structure through the superposition, possessing the advantages of smooth surface, high manufacturing precision, and so on.

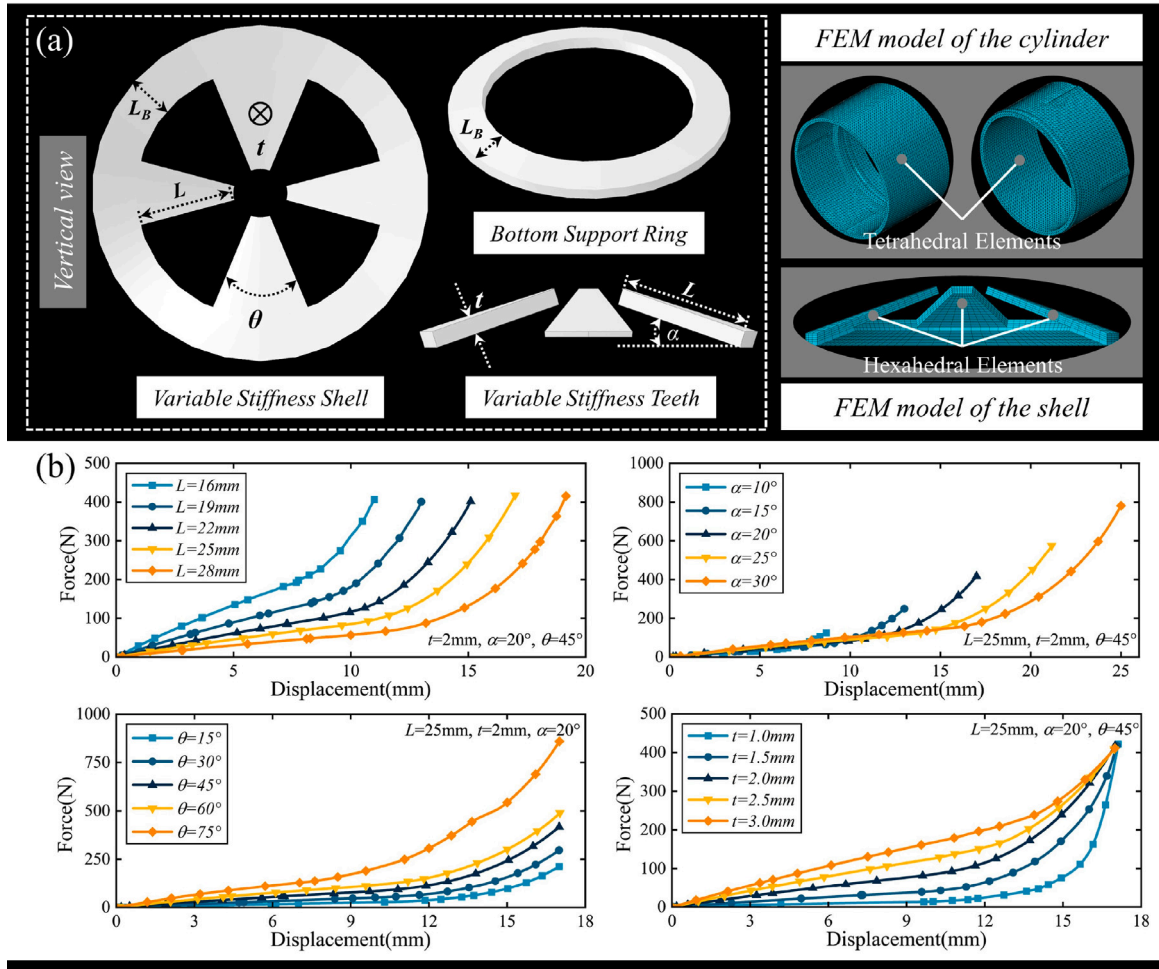


Fig. 2. (a) Structural parameters of the MT-MM unit cell. (b) Force-displacement curves of variable stiffness unit cells with different geometrical parameters.

To investigate the quasi-static compression behaviour of metamaterials, quasi-static compression experiments are carried out on the structure in the vertical direction using the INSTRON 3344 electronic universal testing machine. The structure is placed horizontally on the lower platen. A displacement load is applied to the upper platen, and the loading rate is set at 1 mm/min. The transducer connected to the upper platen would record the displacement and force data during the whole compression process, and plotting force-displacement response curves in real time on the computer.

3. Results and discussion

3.1. Influence pattern of parameters

As depicted in Fig. 2(a), the structural parameters of the MT-MM are given, including the tilt angle of the bottom support ring α , the width of it L_B , the length of variable stiffness teeth L , the thickness t , and the angle of teeth θ . With a view to gaining the influence pattern of different geometrical parameters on the compression characteristics of the variable stiffness unit cells, the investigation of quasi-static mechanical properties is developed by the finite element software ABAQUS. The default structural parameters are configured as follows: $L = 25$ mm, $t = 2$ mm, $\theta = 45^\circ$, $\alpha = 20^\circ$. The compressive force-displacement curves of the structure for different teeth lengths L are displayed in Fig. 2(b), with the increasing of teeth lengths, the force value under the same deformation displacement decreases gradually, implying a decline in the stiffness and the structure is getting flexible. While the tilt angle of the bottom support ring α is increased from 10° to 30° , it

can be observed that the linear segments of force-displacement curves generally overlap, indicating that changing α within a certain range has a minor impact on the stiffness of the structure.

Moreover, as the angle of variable stiffness teeth θ and the thickness t is increasing, the structure gradually becomes 'stiffer' and the bearing capacity is strengthened. It is worth remarking that upon reaching the compression endpoint, it can be observed that the compression force of each structure is almost identical under different teeth lengths L and thicknesses t , but increasing with the increment of tilt angle of support ring α and teeth angle θ . Owing to this, before variable stiffness teeth reach the compression endpoint, its deformation mode is mainly based on the contact between different teeth, at this time the stiffness of the teeth takes the main role. However, at the end of compression, the support ring would be partially deformed, then the overlapping area between the teeth and the support ring would play a major role, demonstrating different loading characteristics according to the contact area of the structure.

3.2. Adjustable stiffness of unit cells

Aiming at further verifying the adjustable properties of MT-MM, as shown in Fig. 3(a), four types of variable stiffness shells with different numbers of teeth pairs and structural parameters are devised respectively, and the specific geometrical parameters are listed in Table 1. The finite element software ABAQUS is used to carry out the investigation of its quasi-static mechanical characteristics. As illustrated in Fig. 3(b), the force-displacement curves at different rotation angles φ are displayed. For a better comparison, the only linear segments of each response

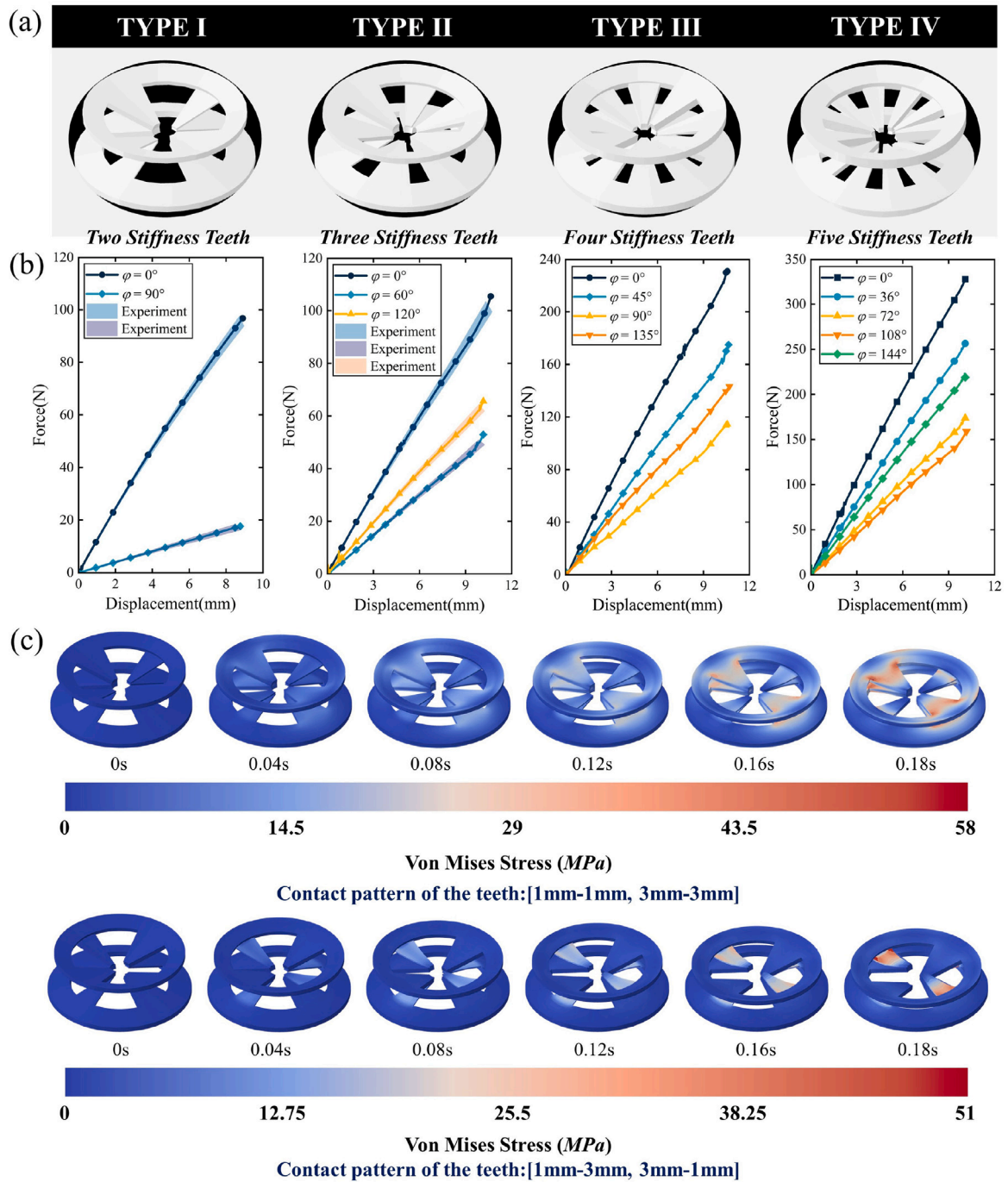


Fig. 3. (a)–(b) Variable stiffness shells with different teeth pairs and structural parameters and as well as force–displacement curves. (c) Deformation process of the structure under two different rotation angles for $N_p = 2$.

curves are taken. It should be noted that, for the variable stiffness shell with the number of teeth pairs N_p , the rotatable angle φ is an integer multiple of $180/N_p$ at each time. As previously mentioned, the limiting portions of internal and external cylinders could be acted as a precise positioning during the rotational adjustment. It could be noticed that the number of stiffness variations exhibited by different teeth pairs of variable stiffness shells is exactly equal to that of N_p . To elaborate further on the variable stiffness mechanism, the deformation process of the structure at two different rotation angles for $N_p = 2$ is exhibited in Fig. 3(c). It could be found that while variable stiffness teeth display the contact mode of ‘1 mm–1 mm, 3 mm–3 mm’, due to the same stiffness of the teeth in contact with each other, all teeth are elastically deformed during compression, showing a larger stiffness. After the structure is

rotated by 90°, the contact mode is changed to ‘1 mm–3 mm, 3 mm–1 mm’, at this time, the stiffness difference of the teeth in contact is much larger, and in the whole compression process, almost only the teeth of 1 mm produce elastic deformation, showing a small stiffness. The difference in stiffness could be more than 5 times.

3.3. Influence of the arrangement

In the previous section, the adjustable stiffness capability of the structure is verified, and this section focuses on the influence of the arrangement on its stiffness characteristics. As depicted in Fig. 4(a), the axial direction along the structure is defined as series and the in-plane

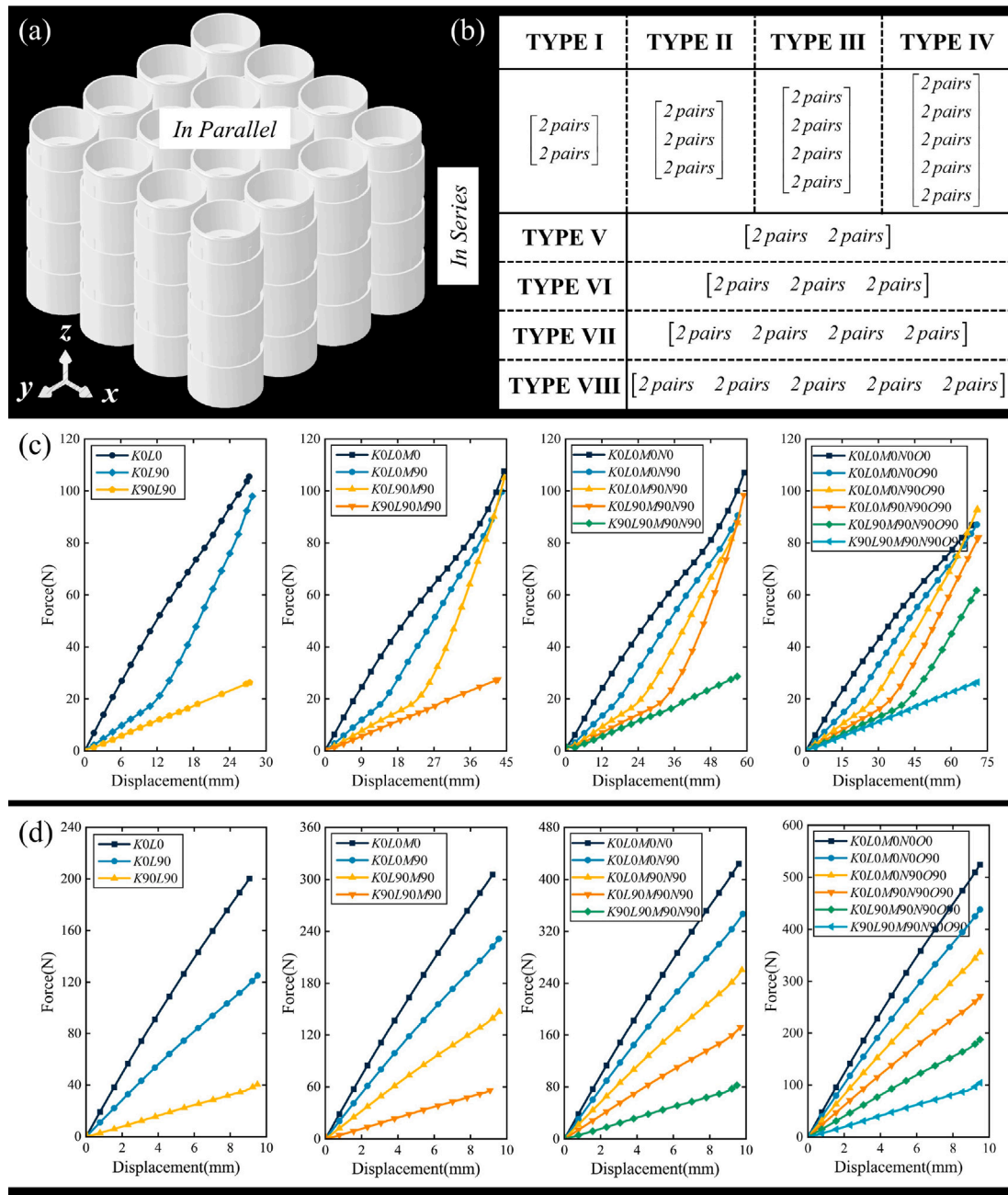


Fig. 4. (a) The arrangement of variable stiffness metamaterials. (b) Different assembly arrangement. (c)–(d) Force–displacement curves of the system: (c) series connection. (d) parallel connection.

Table 1
Structural parameters of variable stiffness shell unit cell.

	N_p	$\theta_{internal}$	$t_{internal}$	$\theta_{external}$	$t_{external}$	φ
TYPEI	2	30/45	1/3	30/45	1/3	90
TYPEII	3	30/35/35	1/2/3	30/35/35	1/1.5/2.5	60
TYPEIII	4	30/30/30/30	1/2/3/4	30/30/30/30	1/1.5/2.5/3.5	45
TYPEIV	5	20/20/20/20/20	1/2/3/4/5	20/20/20/20/20	1/1.5/2.5/3.5/4.5	36

arrangement is considered as parallel. To facilitate the description, the form of the assembly is represented by a matrix. Where, the columns of matrix denote the series connection, the rows represent the parallel, and the matrix value indicates the number of teeth pairs of the variable stiffness shell. For example, [2pairs 2pairs] means connecting two variable stiffness shells with $N_p = 2$ in parallel, and [3pairs 3pairs]^T means connecting three variable stiffness shells with $N_p = 3$ in series.

As depicted in Fig. 4(b), eight different arrangements are provided, where TYPE I-IV denote series connection and TYPE V-VIII denote parallel. For describing the rotation angle of different layers, *KLMNO* is introduced as the number of layer order, and the number of layers is defined as N_{layer} . In particular, it should be noted that φ refers to the rotation angle of the unit cell in the MT-MM, whereas the letters *K, L, M*, etc. represent the angle of rotation for each layer of the combined series or parallel structures. An illustration of force–displacement curves of the structure at different rotational angle arrangements under different series patterns is exhibited in Fig. 4(c). It could be noticed that the global stiffness of the system tends to be weakened with the increase number of layers in series, matching well with the spring-in-series principle. As illustrated in Fig. 4(d), while arranging the variable stiffness shells in parallel, the system stiffness is equal to the sum of the stiffnesses of each layer, improving with the increase of the number of parallel layers. Interestingly, as depicted in Fig. 4(c), the force–displacement curves of the system at certain rotation angles in series arrangement would exhibit a ‘multistep’ behaviour, e.g., *KOL90*, *KOLOM90*, *KOL90M90*, *KOLOM90N90*, and so on. This phenomenon is caused by the significant stiffness difference among different layers of the system, and the behaviour is no longer observed while the stiffness difference is smaller.

4. Conclusions

With the aim of decoupling and adjusting the stiffness, a novel MT-MM is presented in this paper, providing excellent tunable stiffness characteristics and ample stiffness variations. By applying different rotation angles, the structure could exhibit different stiffness properties and achieve stiffness changes from extremely flexible to stiff. For the arrangement of multi unit cells, the system has the stiffness variation feature similar to that of spring series-parallel connection. At present, the MT-MM proposed in this paper is a conceptual investigation. In the future, we will be working on the design and development of adaptive adjustable robotic arms, programmable logic metamaterials and multi-functional tunable bandgap structures.

CRedit authorship contribution statement

Xin Liu: Writing – original draft, Validation, Software, Conceptualization, Methodology, Writing – review & editing. **Shuai Chen:** Conceptualization, Methodology, Software. **Bing Wang:** Writing – review & editing, Project administration, Funding acquisition. **Xiaojun Tan:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

References

- [1] Q. Lin, J. Zhou, K. Wang, D. Xu, G. Wen, Q. Wang, Three-dimensional quasi-zero-stiffness metamaterial for low-frequency and wide complete band gap, *Compos. Struct.* 307 (2023) 116656.
- [2] C. Zeng, L. Liu, C. Lin, X. Xin, Y. Liu, J. Leng, 4D printed continuous fiber reinforced shape memory polymer composites with enhanced mechanical properties and shape memory effects, *Compos. Part A: Appl. Sci. Manuf.* 180 (2024) 108085.
- [3] W. Gao, J. Kang, G. Wang, H. Ma, X. Chen, M. Kadic, V. Laude, H. Tan, Y. Wang, Programmable and variable-stiffness robotic skins for pneumatic actuation, *Adv. Intell. Syst.* 5 (12) (2023) 2300285.
- [4] R.J. Moat, D.J. Clarke, F. Carter, D. Rust, I. Jowers, A class of aperiodic honeycombs with tuneable mechanical properties, *Appl. Mater. Today* 37 (2024) 102127.
- [5] X. Zheng, K. Uto, W.-H. Hu, T.-T. Chen, M. Naito, I. Watanabe, Reprogrammable flexible mechanical metamaterials, *Appl. Mater. Today* 29 (2022) 101662.
- [6] X. Tan, B. Cao, X. Liu, S. Zhu, S. Chen, M. Kadic, B. Wang, Negative stiffness mechanical metamaterials: a review, *Smart Mater. Struct.* 34 (1) (2024).
- [7] T. Mukhopadhyay, J. Ma, H. Feng, D. Hou, J.M. Gattas, Y. Chen, Z. You, Programmable stiffness and shape modulation in origami materials: Emergence of a distant actuation feature, *Appl. Mater. Today* 19 (2020) 100537.
- [8] W. Sun, K. Zhong, Y. Liu, H. Xiao, D. Zhao, Z. Yan, T. Tan, Enhanced metamaterial vibration for high-performance acoustic piezoelectric energy harvesting, *Compos. Commun.* 35 (2022) 101342.
- [9] C. Xiao, R. Tian, X. Zhang, S. Li, Variable stiffness and zero Poisson's ratio of the butterfly-shaped mechanical metamaterial, *Compos. Commun.* (2024) 101958.
- [10] X. Liu, S. Chen, B. Wang, X. Tan, B. Cao, L. Yu, A mechanical metamaterial with real-time tunable bandgap based on pneumatic actuation, *Int. J. Mech. Sci.* (2025) 110045.
- [11] X. An, G. Sun, X. Yuan, Y. Tian, X. Hou, H. Fan, Design of kagome lattice composite sandwich metastructures with high load bearing and low frequency vibration reduction properties, *Compos. Part A: Appl. Sci. Manuf.* 174 (2023) 107716.
- [12] S. Han, N. Ma, H. Zheng, Q. Han, C. Li, Intelligently optimized arch-honeycomb metamaterial with superior bandgap and impact mitigation capacity, *Compos. Part A: Appl. Sci. Manuf.* (2024) 108298.
- [13] R. Xu, Y. He, X. Li, M. Lu, Y. Chen, Snap-fit mechanical metamaterials, *Appl. Mater. Today* 30 (2023) 101714.
- [14] W. Liu, L. Wu, J. Sun, J. Zhou, Origami-inspired quasi-zero stiffness metamaterials for low-frequency multi-direction vibration isolation, *Appl. Phys. Lett.* 123 (8) (2023).
- [15] X. Tan, S. Zhu, B. Wang, K. Yao, S. Chen, P. Xu, L. Wang, Y. Sun, Mechanical response of negative stiffness truncated-conical shell systems: experiment, numerical simulation and empirical model, *Compos. Part B: Eng.* 188 (2020) 107898.
- [16] X. Liu, S. Chen, B. Wang, X. Tan, L. Yu, A compact quasi-zero-stiffness mechanical metamaterial based on truncated conical shells, *Int. J. Mech. Sci.* 277 (2024) 109390.
- [17] H. Yang, J. Zhang, J. Wang, J. Hu, Z. Wu, F. Pan, J. Wu, Delocalized deformation enhanced reusable energy absorption metamaterials based on bistable tensegrity, *Adv. Funct. Mater.* (2024) 2410217.
- [18] Y. Wei, F. Pan, X. Lin, L. Zhang, J. Xiang, Y. Chen, On-demand reprogrammable mechanical metamaterial driven by structure performance relations, *Adv. Mater. (Deerfield Beach, Fla.)* (2024) e2410865.